Roll No. □□□□□□□□□□□□ **Total No. of Pages : 02**

Total No. of Questions : 09

MCA (2015 & Onward) (Sem.–5)

Artificial Intelligence

Subject Code : MCA-501

M.Code : 74381

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTIONS-A, B, C & D contains TWO questions each carrying TEN marks each and students has to attempt any ONE question from each SECTION.**
2. **SECTION-E is COMPULSORY consisting of TEN questions carrying TWENTY marks in all.**

SECTION-A

1. What is an AI technique? Give some examples of problems to explain AI techniques. Why do we need to model human performance in AI?

2. What are Production Systems? Discuss in detail the characteristics of production systems. What relationships are there between problem types and the types of production systems best suited to solve the problems?

SECTION-B

3. Define and distinguish between uninformed search and informed search. Explain the advantages and disadvantages of each. Also discuss one technique of each type.

4. Discuss Game playing as a challenging area of AI. Explain the Alpha-Beta pruning with an example.

SECTION-C

5. What are the desired characteristics of a knowledge representation scheme? Explain the various issues in knowledge representation.

6. What is a Clause? Discuss the procedure to convert a sentence into clausal form with an example.

SECTION-D

7. What do you mean by Natural Language Processing? What are the features of natural languages that create challenges for processing of natural language by computers?

8. Create a Frame network for motor vehicles (motorcycles, cars, trucks) and give one complete frame in detail for cars which includes the slots for the main component parts, their attributes, and relations between parts. Include a Demon slot for the gas of each type mileage.

SECTION-E

9. Answer the following in brief:

   A. Briefly explain the Water Jug problem.

   B. Discuss Turing test.

   C. Differentiate between procedural and declarative knowledge.

   D. What is top-down parsing?

   E. What are difficulties in knowledge acquisition?

   F. What is Inferencing?

   G. Differentiate between a Frame and a Script.

   H. What is Resolution? Briefly explain the resolution principle?

   I. What are quantifiers in predicate logic?

   J. What is pragmatic processing in NLP?

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.